Core2350B ↔ Tang Nano20K · 16-signal wiring map

Core2350B: RP2350B / 8 MB PSRAM / 16 MB flash • Tang Nano20K hardware 3921

FPGA pad wrapper implemented; RP driver / electrical timing unqualified
RP wiring 2026-11-06; 3.3V logic; VGA DAC disconnected

CORE2350B			CABLE / SIGNAL FLOW		TANG NANO20K		
GPIO	Header pin	Signal	At Core	At Tang	Header pin	FPGA pin	Bank
GP39	P4.12	WREADY	33Ω		J6.1	73	1
GP40	P4.11	RVALID	33Ω		J6.2	74	1
GP35	P4.8	ABORT	33Ω		J5.17	72	1
GP36	P4.7	GPU_RESET_N	33Ω		J5.18	71	1
GP37	P4.10	ALE_N	33Ω		J6.15	17	6
GP38	P4.9	RD_N	33Ω		J6.7	16	6
GP16	P3.4	HCLK	33Ω		J6.5	77	1
GP17	P3.3	WR_N	33Ω		J6.6	15	6
GP18	P3.6	D0	33Ω	33Ω	J6.8	27	5
GP19	P3.5	D1	33Ω	33Ω	J6.9	28	5
GP20	P3.8	D2	33Ω	33Ω	J6.12	29	5
GP21	P3.7	D3	33Ω	33Ω	J6.13	30	5
GP22	P3.10	D4	33Ω	33Ω	J6.14	31	5
GP23	P3.9	D5	33Ω	33Ω	J5.5	42	4
GP24	P3.12	D6	33Ω	33Ω	J5.6	41	4
GP25	P3.11	D7	33Ω	33Ω	J5.10	48	3

Power and ground

Power boards through their normal inputs.
Join common GND: Tang J6.20, J5.2, J5.15 to Core P1.1 / P3.1 (fan-out as needed).
Keep inter-board 3.3V and 5V rails separate.
Tang J6.19: local pull-up rail only.
Tang J5.1 (5V), J5.16 (3.3V): unconnected.

Local bias at Tang · 4.7kΩ

To Tang 3.3V: WR_N, RD_N, ALE_N.
To GND: HCLK, ABORT, GPU_RESET_N, WREADY and RVALID. No data pull resistors.
GPU_RESET_N low holds the GPU in reset.
ABORT is active high; *_N is active low.
Pulls attach on the Tang pin side of series R.

Placement and shared pads

33Ω at each driving end; data: 33Ω each end (66Ω total). Starting values; timing unqualified.
Disconnect the Tang FPC LCD and VGA DAC.
Preserve HDMI CEC/HPD (FPGA pins 25/26).
3 loads (15/16/17): orange LED + 510Ω to 3.3V.
Future RP drive ≥8mA per net (12mA advised).

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
Assembly while unpowered is OK; keep removable links open on powered old images. 33Ω does not make contention safe.

Core bus: P3/P4 contacts, not package pins. GP39 WREADY: RP input; shared red LED + 470Ω to GND adds Tang output load.

Sources: user wiring revision; Core2350B schematic; Tang Rev 1.30 / hardware 3921 (2023-07-11). Exact evidence: pinout JSON.

Tang Nano20K · physical header map

Hardware 3921 · component side · USB-C TOP / HDMI BOTTOM

FPGA pad wrapper implemented; RP driver / electrical timing unqualified
RP wiring 2026-11-06; 3.3V logic; VGA DAC disconnected

J6 = LEFT edge; J5 = RIGHT edge. Both number 1 at top → 20 at bottom.



Blue: Core → Tang • Teal: Tang → Core • Purple: bidirectional data
Grey GPIOs: leave bus wiring off. Orange supply pins: never connect to a Core power rail.
Common ground: J6.20 + J5.2 + J5.15 → Core P1.1 / P3.1. Power each board separately.
FPC LCD disconnected. HDMI CEC/HPD (J6.10/J6.11) remain reserved for HDMI.
Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
33Ω does not make contention safe. With old images powered, keep removable links open.

Physical orientation: view the components; do not mirror this map for the solder side. Board outline is schematic, not to scale.

Source: Tang Nano20K Rev 1.30 / hardware 3921 (2023-07-11); full provenance and signal map in the pinout JSON.

Core2350B · P4 / P3 physical locator

RP2350B · 8 MB PSRAM / 16 MB flash · component side · USB FPC connector toward TOP

FPGA pad wrapper implemented; RP driver / electrical timing unqualified
RP wiring 2026-11-06; 3.3V logic; VGA DAC disconnected

Use the Core2350B connector labels below; Pico 2 header numbering does not apply.

P4 · RIGHT EDGE · vertical 2×8 header

Inner / odd		Outer / even	
P4.1 · GND Common ground to Tang	1	2	**P4.2 · GP30** Unallocated by this bus layout
P4.3 · GP32 Unallocated by this bus layout	3	4	**P4.4 · GP31** Unallocated by this bus layout
P4.5 · GP34 Unallocated by this bus layout	5	6	**P4.6 · GP33** Unallocated by this bus layout
P4.7 · GP36 GPU_RESET_N · Tang J5.18	7	8	**P4.8 · GP35** ABORT · Tang J5.17
P4.9 · GP38 RD_N · Tang J6.7	9	10	**P4.10 · GP37** ALE_N · Tang J6.15
P4.11 · GP40 RVALID · Tang J6.2	11	12	**P4.12 · GP39** WREADY · Tang J6.1 · Core LED load
P4.13 · GP42 Unallocated by this bus layout	13	14	**P4.14 · GP41** Unallocated by this bus layout
P4.15 · GP44 Unallocated by this bus layout	15	16	**P4.16 · GP43** Unallocated by this bus layout

P4 contacts read TOP → BOTTOM. Outer column is toward the board's right edge.

GP39: WREADY input only; red LED + 470Ω to GND loads the Tang output. Disable RP LED writes.

P3 · BOTTOM EDGE · horizontal 2×8 header

Read both rows LEFT → RIGHT. Inner / even row is nearer the board centre.

P3.1	P3.2	P3.3	P3.4	P3.5	P3.6	P3.7	P3.8	P3.9	P3.10	P3.11	P3.12	P3.13	P3.14	P3.15	P3.16
GND	GP15	GP17	GP16	GP19	GP18	GP21	GP20	GP23	GP22	GP25	GP24	GP27	GP26	GP29	GP28
Common GND	Unavailable	WR_N	HCLK	D1	D0	D3	D2	D5	D4	D7	D6	PS/2 kept	Unallocated	Unallocated	PS/2 kept
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

INNER / EVEN

OUTER / ODD

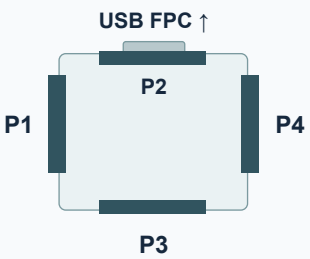
Physical orientation reference

Component side, USB FPC toward top:

P1 left · P2 top · P3 bottom · P4 right.

Signals use P3/P4. Bond P1.1 GND to Tang J5.2 as well.

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed; old powered images need links open.
33Ω does not make contention safe. Power boards separately.



Connector locator only. P3/P4 drawings are expanded for legibility and are not a scale board drawing. Do not mirror for solder side.

Sources: user wiring revision, Core2350B schematic and component-side photograph; provenance in the pinout JSON.